

Nihar Gaddam

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EDUCATION

Carnegie Mellon University

Master of Science in Computer Engineering – Applied Advanced Study

Pittsburgh, PA

Dec 2023

- GPA: 3.75/4.00
- Relevant Coursework: Distributed Systems, Cloud Computing, Foundations of Computer Systems, Computer Vision, Machine Learning for Signal Processing, Introduction to Deep Learning

Sreenidhi Institute of Science and Technology

Bachelor of Technology in Electronics and Communication Engineering

Hyderabad, India

June 2020

- GPA: 9.39/10.00
- Minor in Computer Science and Engineering

SKILLS

Languages: Go, Rust, Python, C++, Java

IAAS: AWS, Azure, GCP

Container Platform: Docker, Kubernetes

CI/CD: GitHub Actions, Apache Airflow

IAC & Config Management: Terraform, Puppet

Storage: HBase, HDFS, Zookeeper, MySQL, MongoDB, Neo4j, Postgres

Frameworks/Tools: PyTorch, Spark, Kafka, Vertex AI

Monitoring and Observability: Prometheus, Grafana, Splunk, OpenTelemetry, SignalFx

EXPERIENCE

Cisco Systems Inc.

Software Development Engineer

Pittsburgh, PA

Aug 2024 – Present

- **Backend Engineering & API Development**
 - Designed and maintained features for a Go-based Kubernetes operator (control plane) managing the full lifecycle of thousands of hosted Splunk Cloud deployments — the isolated, multi-tenant product instances used by customers and internal teams — plus a customer-facing configuration API.
- **System Monitoring & Performance Optimization**
 - Held 24/7 on-call across 5 Kubernetes clusters and 8 services plus the control plane, triaging and resolving production incidents on page while reducing technical debt through targeted refactors across development, staging, and production.
- **Key Projects and Achievements**
 - Built a declarative, self-serve workflow to replace months-long manual provisioning of preview deployments (short-lived demo instances teams trial before adopting the official version): requesters pick preview vs. production and the release version, with validations that strictly isolate the two so they can never be interchanged. Scaled to ~100 preview deployments/month with zero manual coordination; authored the design proposal and drove adoption across five teams.
 - Built the platform's first metrics + SLO observability layer (Prometheus, OpenTelemetry), instrumenting per-phase latency and success/failure metrics across every provisioning stage — infrastructure, software, and final delivery — with data-driven SLOs; shipped pageable alerts that catch genuine regressions proactively and cut cross-team mispages by 30%.
 - Diagnosed and fixed a disaster-recovery teardown ordering bug that could destroy a live primary deployment while its replica was still active; redesigned the control loop to enforce dependency-safe teardown, preventing data loss and eliminating the manual-cleanup incident tickets and on-call pages those mis-ordered teardowns had generated.
 - Built an automated garbage-collection system for stale infrastructure, reclaiming 2,385 abandoned deployments and cutting unnecessary cloud spend fleet-wide.
 - Automated service-to-service authentication with ephemeral credentials, removing 140+ manual tickets/year, and drove encryption-at-rest to 100% coverage across 3,000+ deployments.
 - Enabled customer read access to standby replicas during regional failover, cutting recovery time 30% for 800+ customers.
 - Closed a network security-group filtering gap, reducing related incidents 24%, and surfaced certificate-rotation status to customers, cutting support tickets 23% for 8,000+ users.
 - Authored a design proposal integrating a downstream pooling service with the control plane to pre-provision Azure subscriptions for deployment networking (otherwise created on demand), raising Azure network provisioning success 38% and cutting provisioning time ~20 min.
 - Automated authentication-token rotation across 15 service accounts, eliminating manual intervention and improving token-management efficiency ~40%.

• Cloud Infrastructure Management

- Migrated client infrastructure from on-premises to AWS and Azure, improving scalability and reducing operational costs by 20%; maintained development and production environments, reducing downtime by up to 40%.
- Executed high-risk production changes with minimal disruption, maintaining a 99.8% success rate in change management.

• Data Analysis and Reporting

- Leveraged Power BI for real-time server performance trend analysis, reducing response time to performance issues by 30%; managed ETL dataflows from Azure SQL, SharePoint, Redshift, MongoDB, and Snowflake into BI reporting pipelines.

PROJECTS**Personal****timberdb** | *Go*

Jun 2026

- Built a time-partitioned, TTL-native LSM storage engine in Go; benchmarked 2.0× Pebble, 5.8× Badger, and 13× bbolt append throughput at 3 allocs/op, with 6.6× faster scans than Badger.

SVID-Exchange | *Go*

Mar 2026

- Built a Zero Trust token exchange service that converts SPIFFE SVIDs into scoped, short-lived ES256 JWTs over mTLS, eliminating static shared secrets for service-to-service authentication.

Saga-Conductor | *Go*

Mar 2026

- Built a lightweight saga-pattern orchestrator for distributed transactions: callers define each step with a forward action and a compensating action, and the conductor executes them in order, automatically rolling back completed steps in reverse on failure.

Academic**Distributed Backend Storage for Multiplayer Environments** | *Go*

Dec 2023

- Created a distributed key-value storage system utilizing replication, partitioning, and concurrency mechanisms to support a globally distributed multiplayer online game, ensuring data consistency and low-latency access across multiple backend servers.

Analysis of Twitter User Graph with Spark | *Scala*

Nov 2023

- Proficiently applied Apache Spark programs on Azure HDInsight to analyze the Twitter social graph; implemented the PageRank algorithm to identify influential users and optimized runtime from 60 to 28 minutes, enhancing performance.

Advanced Resource Scaling using Cloud SDKs | *Python*

Sep 2023

- Implemented horizontal scaling with the Boto3 AWS SDK, reducing response times from 30 to 8 minutes; increased throughput from 8K to 40K RPS; configured Elastic Load Balancers, Auto Scaling Groups, and Terraform for efficient deployment.

CERTIFICATIONS**Certified Kubernetes Application Developer (CKAD)**  — Cloud Native Computing Foundation (CNCF) 2024